An Ethical Review of the Pandai Learning App

COMP3041 Professional Ethics · Group Coursework Part 1: Reflective Proposal

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1. Introduction and System Selection

Pandai is a Malaysian learning app, developed by Pandai Education Sdn Bhd, that offers quizzes, topical tests and exam practice aligned with the national curriculum (Pandai Education Sdn Bhd, n.d.). It is designed for primary and secondary school students (MIT Solve, n.d.) and has recently introduced AI-driven features that personalise quizzes and learning paths for each student (Pandai Education Sdn Bhd, n.d.).

We chose Pandai for three reasons. First, it is real and freely available, so every group member could create student and parent test accounts and investigate it. Secondly, it is local and niche, serving Malaysian Year 1 to Form 5 students rather than a global audience, which makes it largely unexplored. Third, it shows several ethically relevant features as it collects personal data from minors, makes automated decisions about what each child studies, handles payments through a Premium subscription, and shapes behaviour through streaks, leaderboards and rewards.

Furthermore, we chose Pandai because its users are children, who understands the commercial use of their data less well than adults (Livingstone et al., 2019). Our proposal examines how far Pandai's design protects its young users, using the Software Engineering Code of Ethics (ACM/IEEE-CS Joint Task Force on Software Engineering Ethics and Professional Practices [ACM/IEEE-CS], 1999) and broader ethical theory.

2. Investigation Methods

We combined four methods. First, we conducted hands-on walkthroughs using student and parent test accounts, including tests of the Ask Pbot chatbot. Second, we analysed Pandai's privacy policy and terms and conditions. Third, we compared the Google Play Data safety section with the privacy policy. Fourth, we checked key screens for dark patterns, such as reward prompts and subscription options. We used only made-up details for all accounts and never entered any real child's information.

3. Literature Review

3.1 Children's privacy and consent

Research shows that children's apps often collect data without meaningful consent. Reyes et al. (2018) analysed nearly 6,000 child-directed Android apps and found that most potentially breached US children's privacy law, often by collecting data before obtaining parental consent. Livingstone et al. (2019) add that children understand commercial uses of their data poorly, so they cannot reliably consent for themselves. However, consent is weak for adults too, Solove (2013) argues that people rarely read privacy policies or foresee the consequences of sharing data, and Obar and Oeldorf-Hirsch (2020) found that most users skip privacy policies entirely. Together, these studies suggest that asking a child to accept terms at sign-up is not meaningful consent, placing responsibility on designers instead.

3.2 AI in education

AI-driven personalisation promises tailored learning but raises ethical concerns that research has only begun to address. Holmes et al. (2022) surveyed leading AI-in-education researchers and found that most had received no formal ethics training, concluding that the field lacks a shared ethical framework. Regan and Jesse (2019) identify the specific risks: personalised learning systems collect detailed data on students, track their behaviour and sort them algorithmically, threatening their privacy and autonomy. UNICEF (2021) goes further, setting out requirements that AI systems used by children must be transparent, explainable and protective of their data. Together, these sources suggest that children's AI tools need stronger safeguards than adult systems.

3.3 Gamification and dark patterns

Gamification is the use of game elements such as points, badges and leaderboards in non-game settings, can motivate learning but also manipulate users. Kim and Werbach (2016) argue that gamification is ethical when it respects users' autonomy and serves their interests, but becomes exploitative when it pressures or deceives them. Gray et al. (2018) show that designers often cross this line, using "dark patterns" that steer users towards choices benefiting the business rather than themselves. Radesky et al. (2022) found such manipulative designs to be common in apps used by children, who are less able to recognise them. Together, these sources suggest that game mechanics in children's apps need scrutiny.

3.4 Privacy labels and transparency

Privacy labels aim to give users a quick, standardised summary of how an app handles their data. Kelley et al. (2009) proposed this "nutrition label" approach, showing that users understood a short, structured label more easily than a full privacy policy. However, later studies question whether labels are accurate. Kollnig et al. (2022) found that many iOS privacy labels did not match the data the apps collected, and Arkalakis et al. (2024) reported widespread inconsistencies in Google Play's Data safety section, partly because developers declare their own practices with little verification. These findings suggest that labels can create a false sense of transparency, misleading the parents who rely on them.

4. Ethical Issues Identified

4.1 Issue 1: Children's data and consent

Claim: Pandai knows its users are children, yet collects their data without parental approval, assuming consent at sign-up and offering no clear way to withdraw it.

Evidence: Our walkthrough found that students sign up with only a name, username and password. Consent is implied by the line "By registering, I accept the Terms and Conditions", with no checkbox, privacy policy link or parent step (Figure 1a). The app only accepts ages 6 to 18 (Figure 1b), so Pandai knows every student is a minor, yet it still asks for their home address to send "gift and rewards" (Figure 1c). Parents can skip linking a child entirely, and linking needs only the child's six-digit PIN, with no proof of guardianship. Once linked, the adult can view the child's activity and switch into their account (Figure 2).

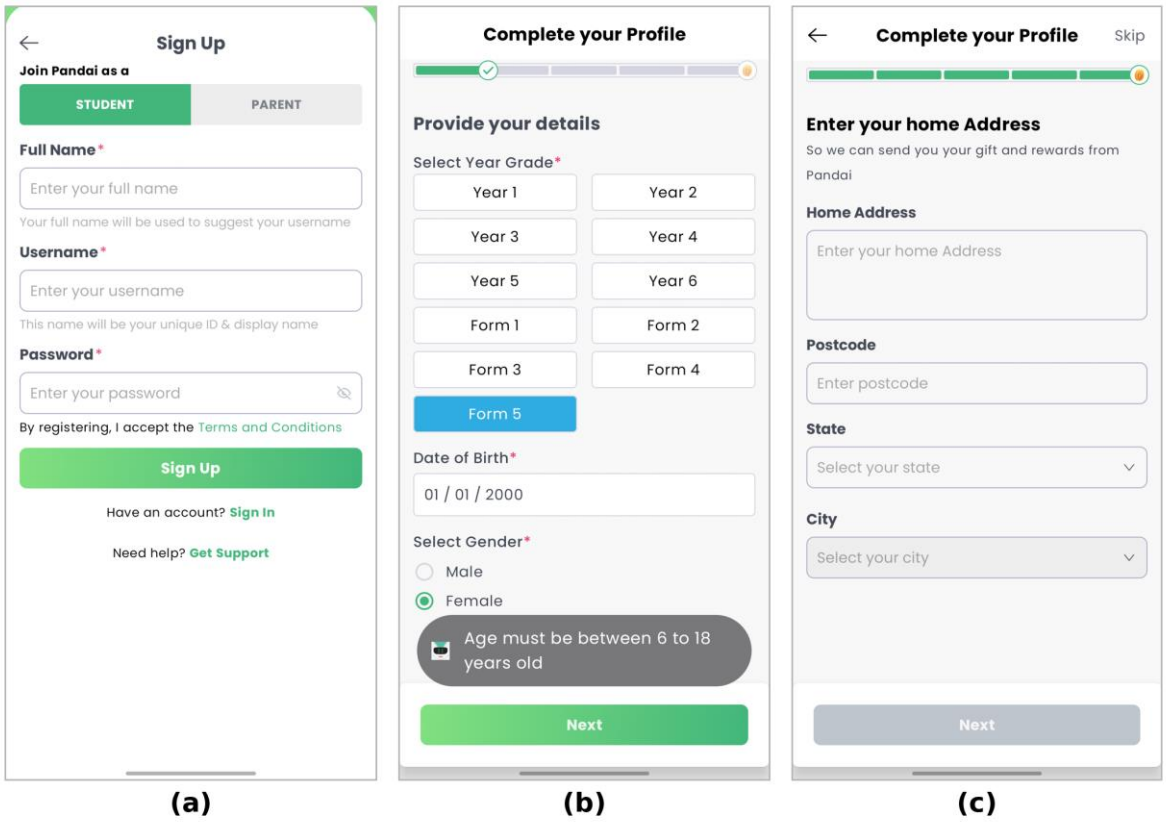


Figure 1: Student sign-up walkthrough. (a) Consent implied by registering; (b) enforced age range of 6–18; (c) home address requested for "gift and rewards".

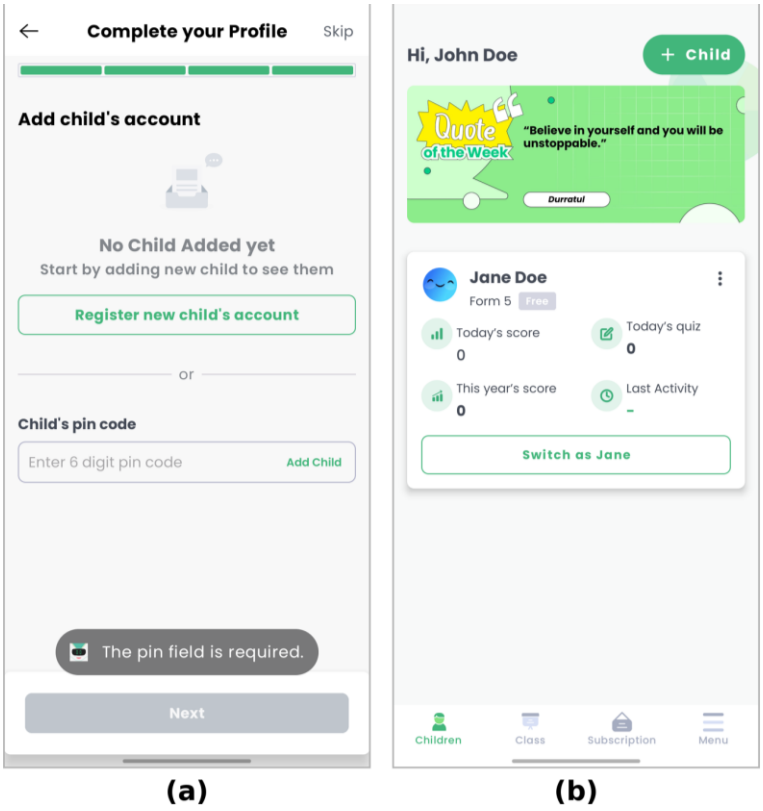
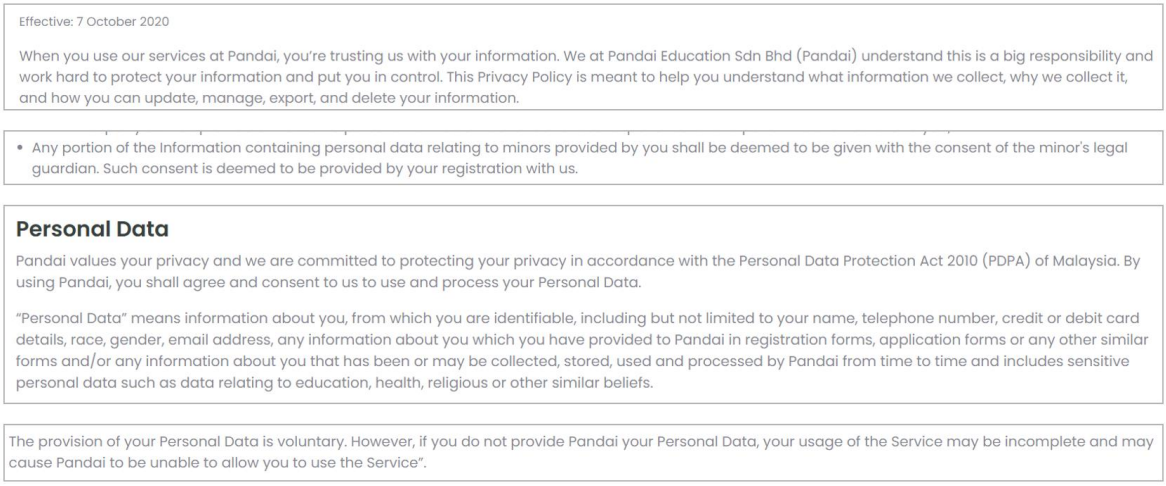


Figure 2: Parent account. (a) Linking a child is skippable and requires only the child's 6-digit PIN; (b) parent dashboard with "Switch as Jane"

Pandai's privacy policy explains this design (Figure 3). It treats registration as automatic guardian consent for minors and use of the app as consent to data processing. It calls providing data "voluntary", yet warns that refusing may block access to the service. The policy also promises that users can export and delete their data but describes no way to do so, and it has not been updated since 7 October 2020 (Pandai Education Sdn Bhd, 2020).



Effective: 7 October 2020

When you use our services at Pandai, you're trusting us with your information. We at Pandai Education Sdn Bhd (Pandai) understand this is a big responsibility and work hard to protect your information and put you in control. This Privacy Policy is meant to help you understand what information we collect, why we collect it, and how you can update, manage, export, and delete your information.

- Any portion of the Information containing personal data relating to minors provided by you shall be deemed to be given with the consent of the minor's legal guardian. Such consent is deemed to be provided by your registration with us.

Personal Data

Pandai values your privacy and we are committed to protecting your privacy in accordance with the Personal Data Protection Act 2010 (PDPA) of Malaysia. By using Pandai, you shall agree and consent to us to use and process your Personal Data.

"Personal Data" means information about you, from which you are identifiable, including but not limited to your name, telephone number, credit or debit card details, race, gender, email address, any information about you which you have provided to Pandai in registration forms, application forms or any other similar forms and/or any information about you that has been or may be collected, stored, used and processed by Pandai from time to time and includes sensitive personal data such as data relating to education, health, religious or other similar beliefs.

The provision of your Personal Data is voluntary. However, if you do not provide Pandai your Personal Data, your usage of the Service may be incomplete and may cause Pandai to be unable to allow you to use the Service".

Figure 3: Pandai privacy policy extracts: effective date and stated user controls; deemed guardian consent; consent by use and definition of personal data; "voluntary" provision of data.

This matches the pattern Reyes et al. (2018) found across children's apps, where data is collected before parental consent, and the weak consent model Solove (2013) criticises. It conflicts with Clauses 1.03

and 3.12 of the SE Code (ACM/IEEE-CS, 1999), which require software that protects users' privacy, and engages Clause 1.06 on avoiding deception, as the promised deletion controls are never described.

4.2 Issue 2: AI chatbot transparency

Claim: Pandai offers children an AI chatbot without explaining how their conversations are stored or used, leaving students and parents unable to make informed choices about sharing their data.
Evidence: Ask Pbot introduces itself as an "AI study buddy" and invites students to "ask me anything" (Figure 4a). When asked what it does with chats, it replied that it does not "know exactly how Pandai stores or uses chats" and directed us to Pandai's privacy information (Figure 4b). However, the privacy policy, in effect since 2020, never mentions AI or chat data (Figure 3). Pbot also answered off-topic questions, such as recommending a film (Figure 4c), suggesting its scope is not limited to learning. In fairness, it solved our algebra and differentiation questions correctly and displays a disclaimer that it "may make mistakes".

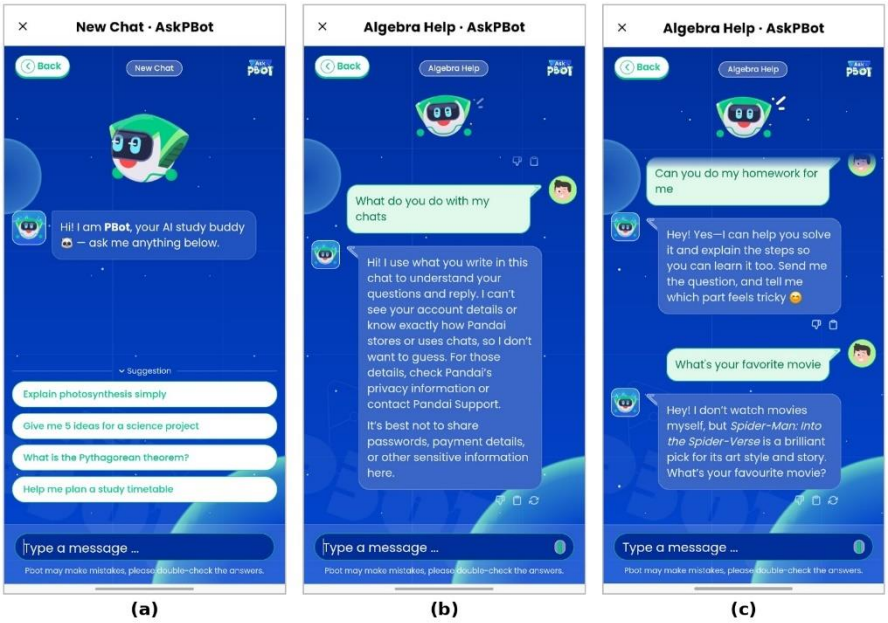


Figure 4: Ask Pbot walkthrough. (a) Opening screen with "may make mistakes" disclaimer; (b) response on chat data use; (c) off-topic answer

This falls short of UNICEF's (2021) call for transparent AI for children and reflects the data risks Regan and Jesse (2019) describe in personalised learning systems. It engages Clauses 1.03 and 1.04 of the SE Code (ACM/IEEE-CS, 1999), which require engineers to protect privacy and disclose potential risks to users.

4.3 Issue 3: Behaviour shaping

Claim: Pandai uses expiring coins, public leaderboards and game-linked rewards that may serve retention and Premium sales as much as learning.

Evidence: Coins expire unless a quiz is submitted within 36 hours, and the prompt offers a subscription to "keep them permanently" (Figure 5a). Starting a new topic costs a coin unless 12 stars are earned (Figure 5b). Coins can be redeemed for gaming eVouchers, such as Mobile Legends, PUBG and Roblox credit, tagged Premium or Premium Lite (Figure 5c). A national leaderboard displays children's names, photos and scores (Figure 6a), and locked badges encourage streaks (Figure 6b). In fairness, rewards depend on completing learning tasks.

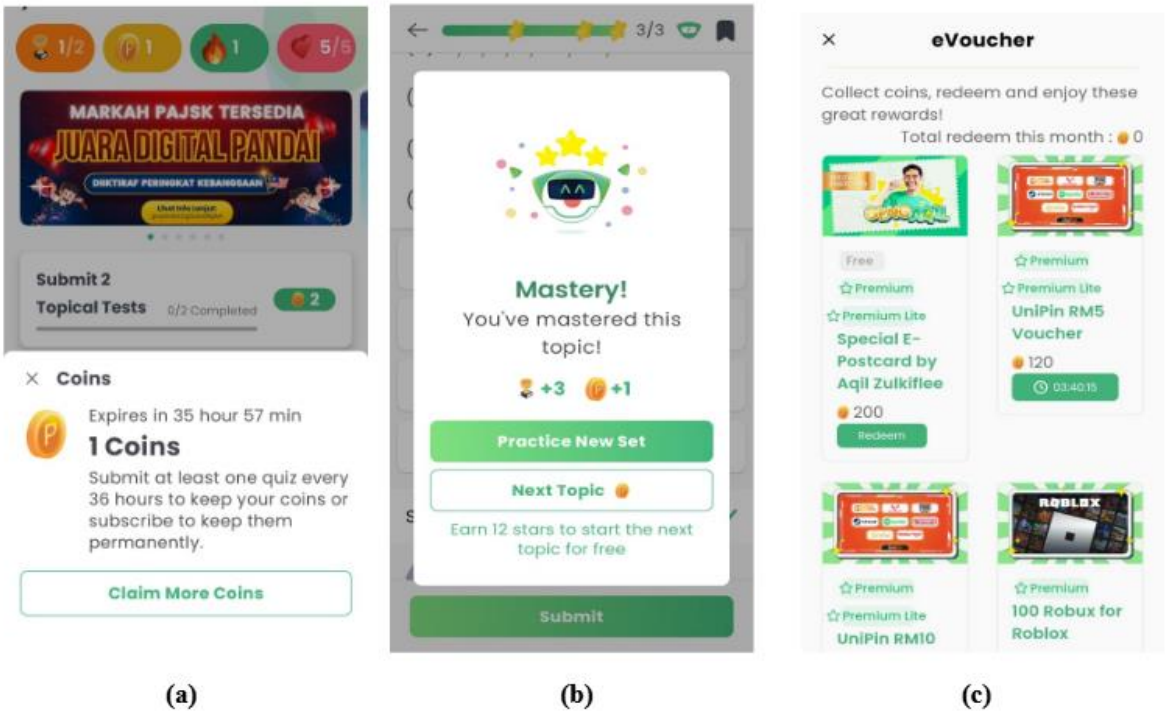


Figure 5: Pandai reward mechanics. (a) Coins expire after 36 hours unless a quiz is submitted; (b) next topic costs a coin unless 12 stars are earned; (c) eVouchers include gaming vouchers tagged Premium/Premium Lite.

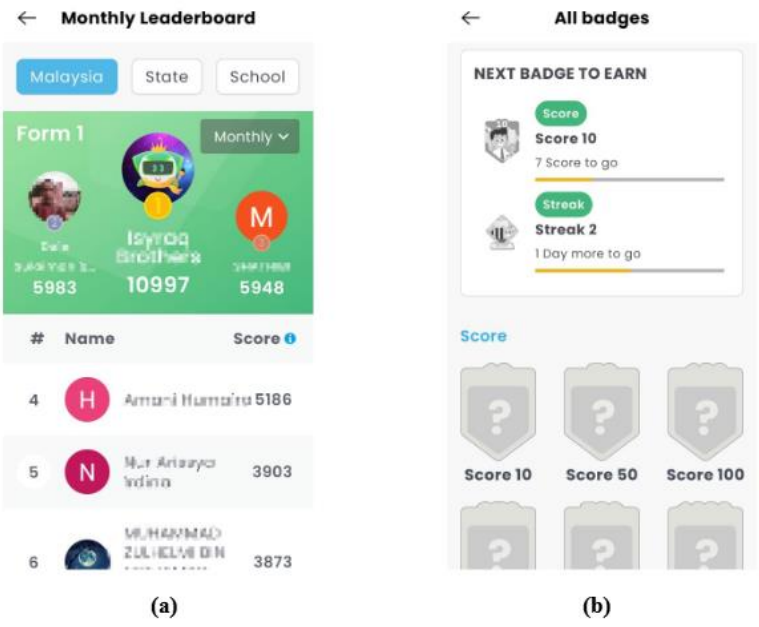


Figure 6: Competition and progress features. (a) National monthly leaderboard (other users' names and photos have been blurred); (b) badge progress with locked badges.

The 36-hour expiry resembles the time pressure Radesky et al. (2022) found in children's apps, and linking coin retention to subscription fits Gray et al.'s (2018) account of design that serves business goals. Under Kim and Werbach's (2016) framework, game-credit rewards and public rankings of children raise manipulation concerns.

This engages Clause 1.03, as loss-framed expiry and public ranking may pressure and expose children, and Clause 1.02, which requires engineers to balance employer interests with the public

good. Under Clause 2.07, engineers building these features should report such concerns to their employer.

4.4 Issue 4: Transparency gap

Claim: Pandai describes its data practices inconsistently across its Google Play label, App Store label and privacy policy, so parents cannot tell from any single source how data is used.

Evidence: Google Play's Data safety section states that no data is shared with third parties (Figure 7), yet the App Store label states that name, email, address and device ID may be used to track users across other companies' apps (Figure 8). The iOS tracking prompt cites a "personalized experience" without mentioning advertising, while the privacy policy permits third-party behavioural analysis but never mentions cross-app tracking (Figure 9). Because Google and Apple define "sharing" and "tracking" differently, these gaps do not prove any declaration false; however, what a parent learns depends on which source they read.

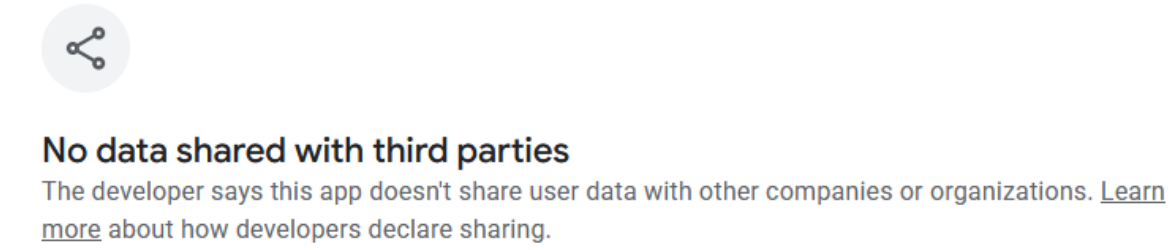


Figure 7: Google Play Data safety label showing no sharing and analytics-only collection

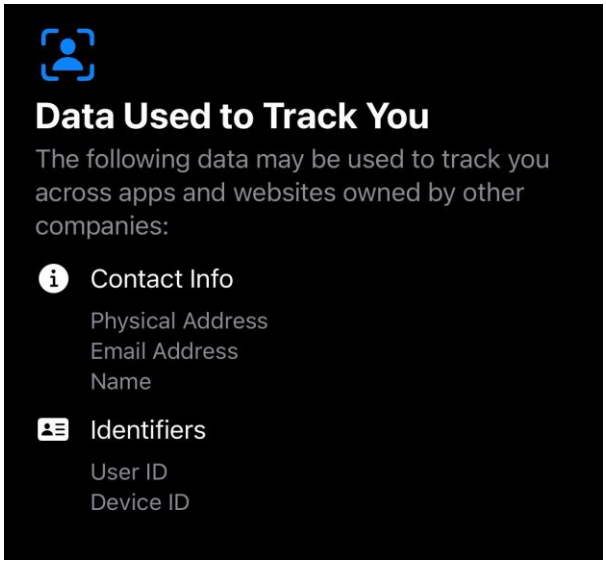
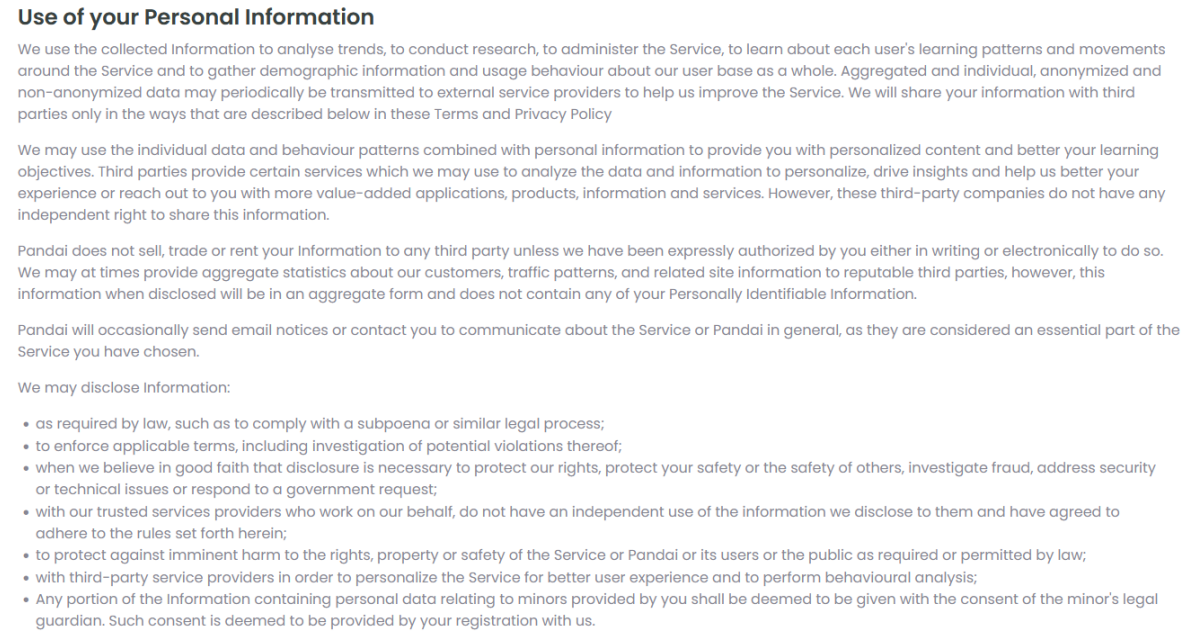


Figure 8: App Store privacy label showing data used to track users.



Use of your Personal Information

We use the collected Information to analyse trends, to conduct research, to administer the Service, to learn about each user's learning patterns and movements around the Service and to gather demographic information and usage behaviour about our user base as a whole. Aggregated and individual, anonymized and non-anonymized data may periodically be transmitted to external service providers to help us improve the Service. We will share your information with third parties only in the ways that are described below in these Terms and Privacy Policy

We may use the individual data and behaviour patterns combined with personal information to provide you with personalized content and better your learning objectives. Third parties provide certain services which we may use to analyze the data and information to personalize, drive insights and help us better your experience or reach out to you with more value-added applications, products, information and services. However, these third-party companies do not have any independent right to share this information.

Pandai does not sell, trade or rent your Information to any third party unless we have been expressly authorized by you either in writing or electronically to do so. We may at times provide aggregate statistics about our customers, traffic patterns, and related site information to reputable third parties, however, this information when disclosed will be in an aggregate form and does not contain any of your Personally Identifiable Information.

Pandai will occasionally send email notices or contact you to communicate about the Service or Pandai in general, as they are considered an essential part of the Service you have chosen.

We may disclose Information:

- as required by law, such as to comply with a subpoena or similar legal process;
- to enforce applicable terms, including investigation of potential violations thereof;
- when we believe in good faith that disclosure is necessary to protect our rights, protect your safety or the safety of others, investigate fraud, address security or technical issues or respond to a government request;
- with our trusted services providers who work on our behalf, do not have an independent use of the information we disclose to them and have agreed to adhere to the rules set forth herein;
- to protect against imminent harm to the rights, property or safety of the Service or Pandai or its users or the public as required or permitted by law;
- with third-party service providers in order to personalize the Service for better user experience and to perform behavioural analysis;
- Any portion of the Information containing personal data relating to minors provided by you shall be deemed to be given with the consent of the minor's legal guardian. Such consent is deemed to be provided by your registration with us.

Figure 9: Pandai privacy policy extracts: "does not sell, trade or rent" statement alongside third-party sharing for personalisation and behavioural analysis (pandai.org/en/privacy-policy).

This reflects Lin et al.'s (2024) finding that Android and iOS labels define data practices differently, confusing users, and adds to the reading burden McDonald and Cranor (2008) quantified. The tracking prompt's omission of advertising illustrates the weak consent Solove (2013) criticises.

This engages Clause 1.06 on non-deceptive public statements and Clause 3.01, which requires significant tradeoffs to be available for users' consideration. Since each platform's form may be completed accurately yet still mislead collectively, the problem lies partly beyond individual engineers.

5. Ethical Framework

5.1 Ethical theories

We apply two complementary theories. Kantian deontology judges actions by whether they respect people as ends in themselves rather than using them merely as means (Kant, 1785/2012). This suits Pandai because its key issues concern consent and respect: deeming children to have consented at registration (Issue 1), and using coins, streaks and game vouchers to drive engagement and subscriptions (Issue 3), risk treating children as means to data and revenue rather than as learners.

Utilitarianism judges actions by their overall consequences, favouring those that produce the greatest good for the greatest number (Mill, 1863/2001). This allows us to weigh Pandai's genuine benefit, free curriculum-aligned practice for many Malaysian students, against the privacy and wellbeing harms identified in Issues 1 to 4. Using both theories prevents a one-sided judgement: deontology identifies duties Pandai may breach, while utilitarianism asks whether its benefits outweigh those harms.

5.2 Where the SE Code is insufficient

The SE Code cannot fully resolve Issue 3. Pandai's coins, streaks and game vouchers may motivate students to practise, serving the client's interests under Principle 2, yet the same features pressure children to keep returning and push families towards Premium, which may diminish their quality of life (Clause 1.03). Clause 1.02 asks engineers to balance these interests with the public good but offers no method for weighing them, and the Code is silent on children as a distinct vulnerable group (ACM/IEEE-CS, 1999). Furthermore, decisions such as linking rewards to subscriptions are made by management rather than engineers, placing them largely outside the Code's scope. Ethical theory is therefore needed: Kantian deontology asks whether children are treated as ends rather than as means to

revenue, while utilitarianism weighs the educational benefit of these features against their wellbeing and financial costs.

6. Conclusion and Next Steps

Our investigation indicates that Pandai's design raises concerns about children's consent, AI transparency, reward-driven engagement and the accuracy of its privacy disclosures. To complete the review, we will first test whether account deletion is possible in the app, to determine whether consent can be withdrawn in practice. We will check whether children are notified when a parent links their account, to assess whether children have any say over parental access. We will also test Ask Pbot on harder questions and examine whether Pandai explains its quiz recommendations, to judge the reliability and transparency of its AI. Finally, we will code a larger sample of user reviews to see whether users themselves report these issues. We will then evaluate each issue against the SE Code clauses identified, applying Kantian deontology and utilitarianism where the Code is insufficient, and propose practical design recommendations.

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